Week 4
Inheritance and Polymorphism
Program 10
Write a Java program demonstrating single inheritance using Person and Student classes.

```
class Person {
   String name;
   int age;

   void displayPerson() {
      System.out.println("Name: " + name);
      System.out.println("Age: " + age);
   }
}

class Student extends Person {
   int rollNo;
   String course;

   void displayStudent() {
      displayPerson();
      System.out.println("Roll Number: " + rollNo);
      System.out.println("Course: " + course);
   }

   public static void main(String[] args) {
      Student s = new Student();

      s.name = "Rahul";
      s.age = 20;
      s.rollNo = 101;
      s.course = "B.Tech CSE";

      s.displayStudent();
   }
}
```

OUTPUT:
Name: Rahul
Age: 20
Roll Number: 101
Course: B.Tech CSE

Program 11
Write a Java program demonstrating multilevel inheritance using Animal, Mammal, and Dog classes.

```
class Animal {
   void eat() {
```

```
        System.out.println("Animal eats food");
    }
}

class Mammal extends Animal {
    void walk() {
        System.out.println("Mammal can walk");
    }
}

class Dog extends Mammal {
    void bark() {
        System.out.println("Dog barks");
    }

    public static void main(String[] args) {
        Dog d = new Dog();

        d.eat();
        d.walk();
        d.bark();
    }
}
```

OUTPUT:
Animal eats food
Mammal can walk
Dog barks

Program 12
Write a Java program illustrating method overriding.

```
class Animal {
    void sound() {
        System.out.println("Animal makes a sound");
    }
}

class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Dog barks");
    }

    public static void main(String[] args) {
        Dog d = new Dog();
        d.sound();
```

```
        Animal a = new Animal();
        a.sound();
    }
}
```

OUTPUT:
Dog barks
Animal makes a sound